

Geotech-BC Borehole Camera

Application

- Check the quality of concrete pouring and the effect of grouting treatment
- Monitoring and detection of geological disasters
- Guide the installation of underground instruments and equipment
- Inspection and detection of underground pipeline and advance detection of tunnel excavation
- Divide lithology, find out geological structure and determine weak argillaceous interlayer
- Detection of faults, fissures and fracture zones
- Observe groundwater activity and assist in Geomechanical test
- Coal well, oil well detection, etc



Features

• 360° Panoramic Imaging

Using 360 ° Panoramic HD ultra low light camera, the front end of the probe adopts an open toughened glass hemispherical cover, which can take a 360 degree omni-directional camera of the drilling wall.

• Ultra HD + true color

Standard 5 megapixel high-definition lens, digital image color realistic, 1:1 imaging, picture amplification without Mosaic, crack resolution up to 0.01mm.

• Stitching pictures+ video Combination

By using efficient intelligent image unfolding algorithm, when the probe is pulled up and down, it can seamlessly and quickly make a mosaic and video recording, and save and play back the whole cylindrical unfolding drawing of the wellbore and the downhole forward-looking video in real time

• Auto Detection

Using stepper motor to lift and lower the probe at a constant speed can avoid the problems of image jitter and depth error caused by manual cable lowering.

• 3D high-definition Sensor

Built in high-precision electronic compass, three-axis sensor, accuracy of up to $\pm$ zero point one °. It can measure all kinds of vertical holes, inclined holes and horizontal holes.

• Software

The electronic core library of hole wall high-definition image data replaces the physical core to solve the problems of core sampling, transportation, storage and management.

Configuration



Geotech-BC Host



Strander with Cable



Counting pulley



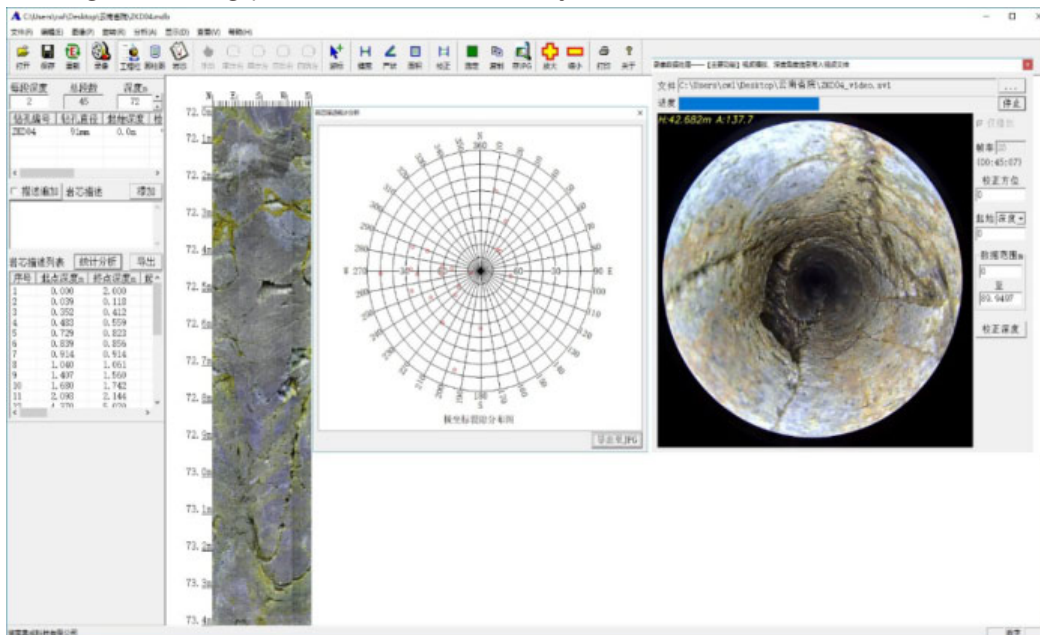
3D HD probe

Specifications

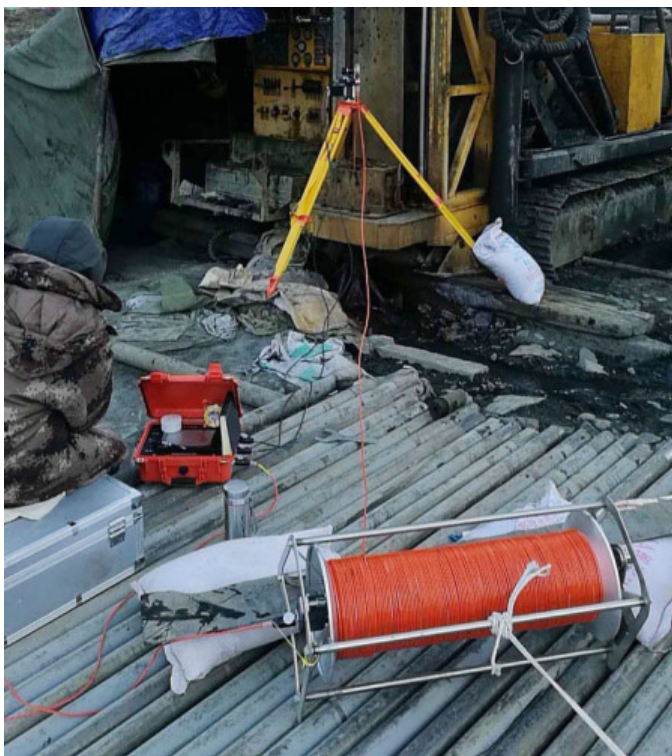
Imaner	Multi-media processing industrial control machine
Operation	Touch screen + button
Display	8.4-inch TFT LCD
Storage	128G high speed solid state hard disk
Collecting speed	Up to 60 frames per second
Output picture	≥2048*2048pix
Collecting mode	real-time image mosaicking, real-time video recording, real-time image mosaicking and video recording
Host size	350*300*150mm
Host weight	3.5KG
The lens type	360° Panoramic lens
Open Camera	The probe camera use hemispherical explosion-proof glass cover, with no shielding in front, and easy to disassemble and replace;
Probe resolution	2-10 megapixels
Probe diameter	Φ50mm, Φ63mm (others on request)
Probe lifting mode	Uniform electric lifting
Probe lifting speed	0.5 ~ 3.0m/min
Length of the cable	100 ~ 1500m(customizable)
Counting accuracy	0.1mm
Open light source*	Open waterproof LED ring light source design, no dead Angle, no gouge, better imaging effect in the hole
3D sensor*	3d attitude sensor is adopted to indicate the attitude of the probe in the well in real time, supporting the detection of vertical hole, horizontal hole and inclined hole
Cable*	Customized cable, 100m cable weight <2kg, tensile resistance more than 200kg;
Sensor accuracy	Crevice accuracy:0.1mm, azimuth accuracy: 0.1°
Battery life	8 Hours

Cases

Drilling Stitching pictures + video analysis interface



Some engineering application sites



Preliminary Survey of Kangding Section of Sichuan-Tibet Railway



Inclined hole inspection of Baihetan Hydropower Station

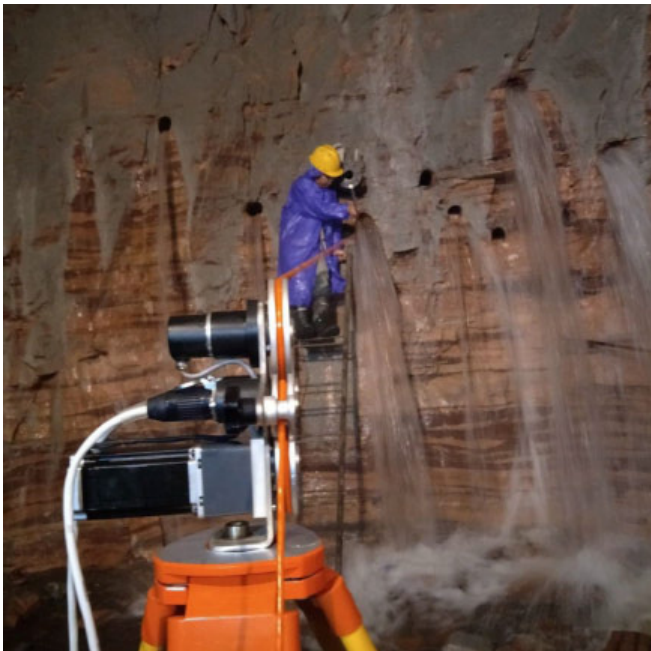
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Guizhou Rail Line 3 Phase Project Inspection



Horizontal hole inspection of Xiaowan Hydropower Station

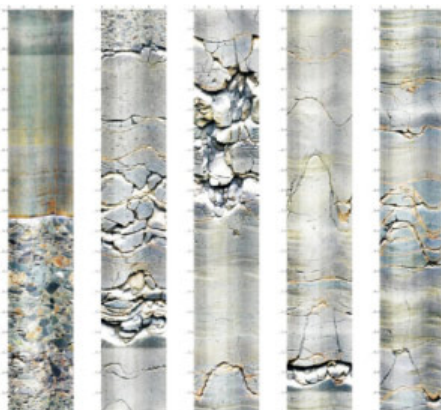


Horizontal hole inspection of tunnel engineering

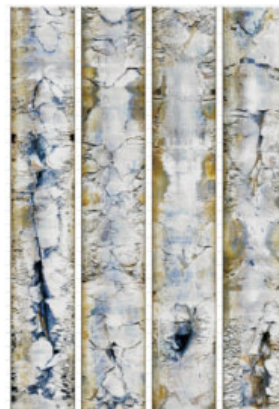


Preliminary survey of a subway project in Nanjing

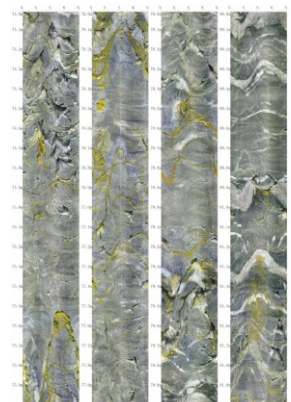
Some borehole measured data



Measured data of 91mm diameter borehole

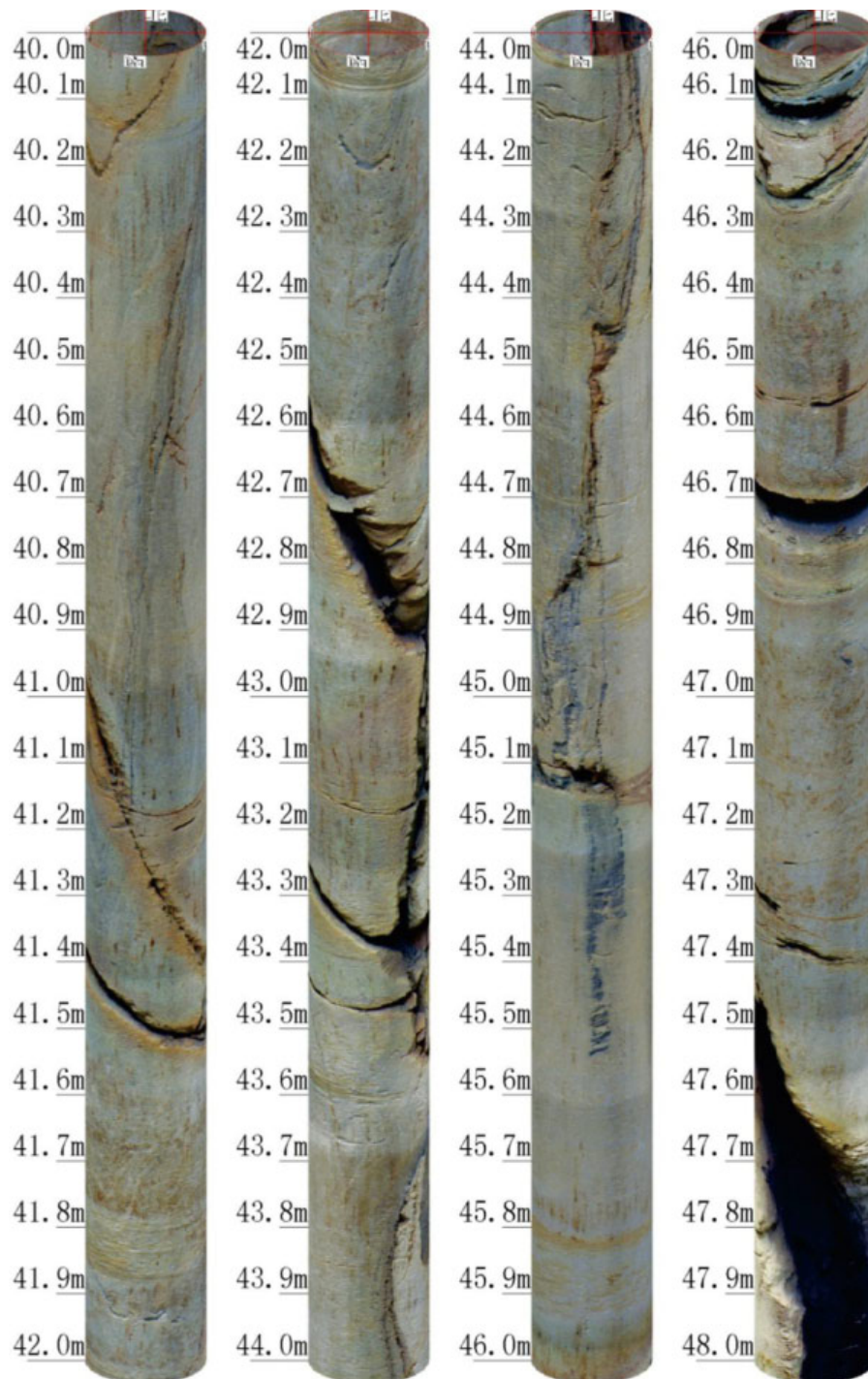


Measured data of 76mm diameter drilling hole



Actual data of 60mm diameter drilling hole

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110mm diameter borehole is measured



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